

Basic Principles of Medicine II
Module: Nervous System and
Behavioral Sciences
Lecture No: NB 04

Meninges and Arteries

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Recommended Reading

1. <https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=196986286&bookid=2482>
2. [Meninges, Cisterns, Ventricles, and Related Hemorrhages | Neuroanatomy Atlas in Clinical Context Structures, Sections, Systems, and Syndromes, 11e | Premium Basic Sciences | Health Library \(lwwhealthlibrary.com\)](#)

Objectives

SOM.MK.I.BPM2.1.NB.1.NSCI.0070 Describe the structural relations among the cranial meninges.

SOM.MK.I.BPM2.1.NB.1.NSCI.0071 Distinguish the four types of intracranial hemorrhages based on blood source and appearance on neuroimage.

SOM.MK.I.BPM2.1.NB.1.NSCI.0072 Identify the major subarachnoid vessels and their general cerebral distributions.

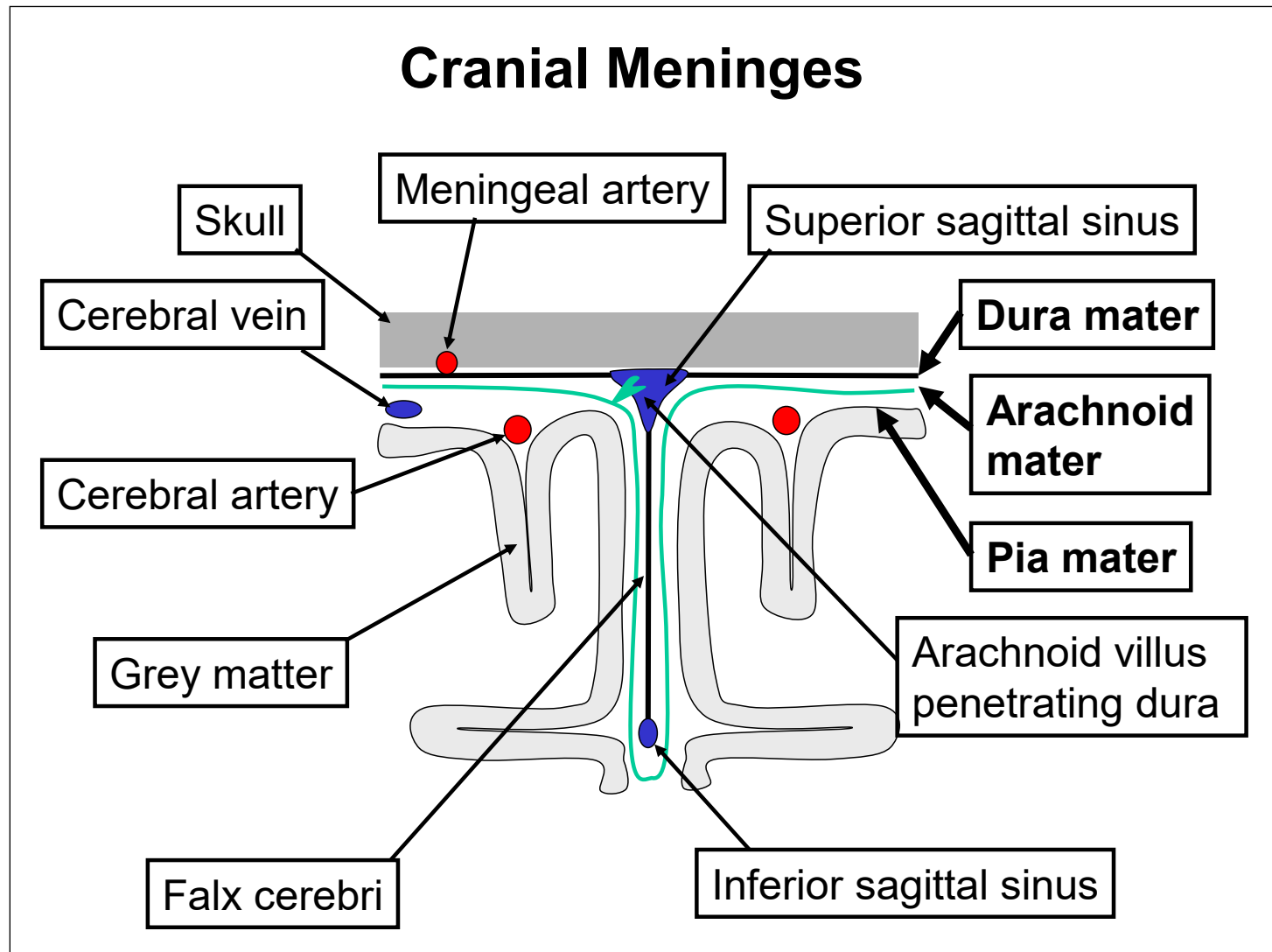
SOM.MK.I.BPM2.1.NB.1.NSCI.0562 Identify the major subarachnoid vessels and their general bulbar distributions.

SOM.MK.I.BPM2.1.NB.1.NSCI.0073 Describe the structural relations among the spinal meninges.

SOM.MK.I.BPM2.1.NB.1.NSCI.0074 Describe the sources and distribution of spinal blood, as depicted in horizontal sections.

- There are four layers of cranial meninges: **Dura mater** (two layers), **arachnoid** and **pia mater**.
- The **periosteal dura mater** adheres to the inner surface of the skull. Arteries supplying the meninges thus dwell against and indent the skull.
- Dural folds (formed by the inner or meningeal layer of the dura) dwell between the major subdivisions of the brain, forming large septa (**falx cerebri**, **tentorium cerebelli**).
- **Dural venous sinuses** (large valveless vessels) often lie between the two layers of dura mater.
- The thin avascular **arachnoid** lies under the dura mater.
- The **pia mater** is highly vascular and covers the outer surface of the brain.

Cranial Meninges



Coronal view near vertex of skull

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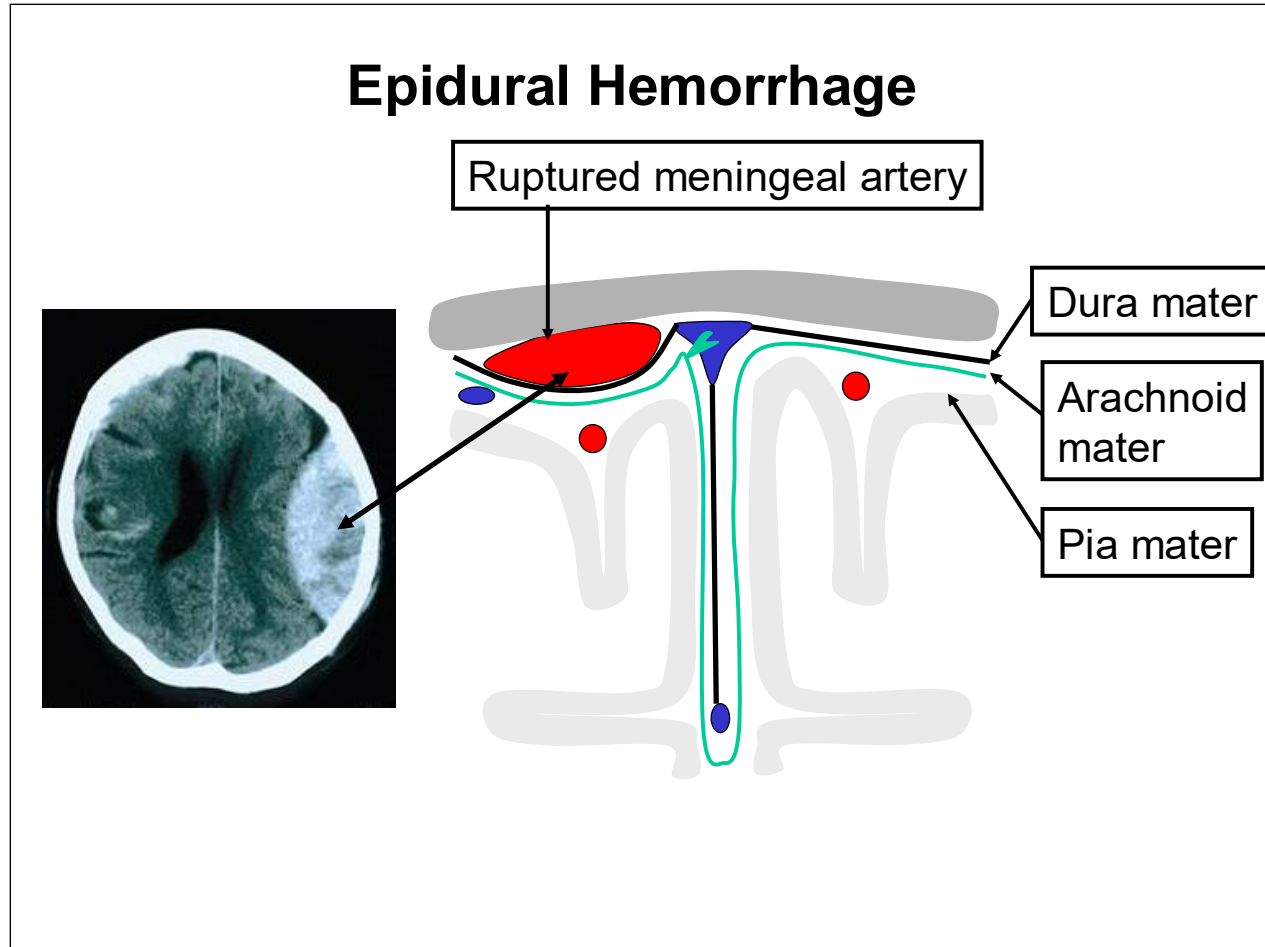
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- Epidural hemorrhages commonly reflect the traumatically induced rupture of meningeal arteries (e.g., **middle meningeal artery**), with a fractured bone cutting the vessel. Leaking blood separates the stiff periosteal dura from the skull, yielding a characteristic biconvex (lenticular; lens-shaped) mass adjacent to the fracture. The hematoma is contained by the penetrations of the periosteal dura into the sutures of the skull. Pressure arising from the accumulating blood compresses/distorts the brain. Dangers typically peak within hours of arterial involvement but may go fatally unrecognized.

Epidural Hemorrhage

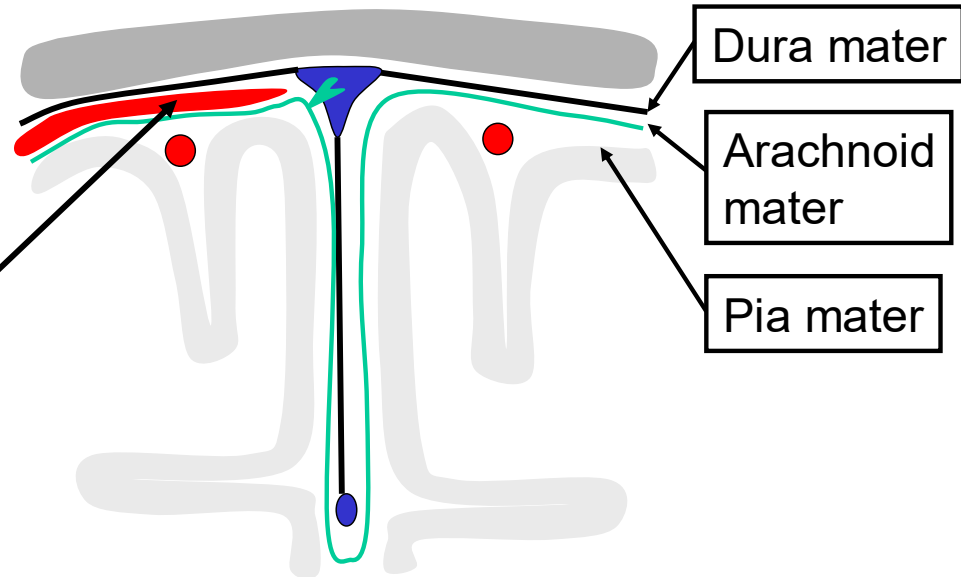
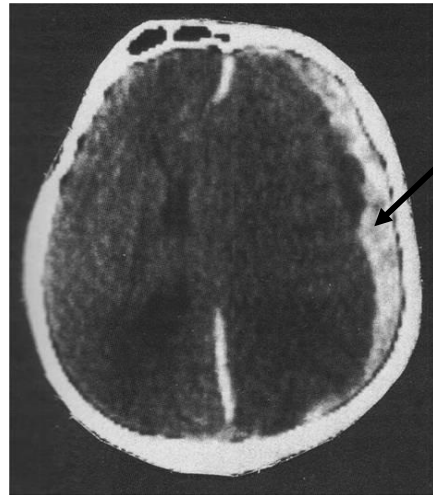


- Note biconvex (lenticular) hematoma

- Subdural hemorrhages commonly reflect trauma-induced ruptures of the delicate veins that traverse the meningeal layer of the dura from the subarachnoid space (bridging veins). Bleeds arising from tearing of the meningeal dura are also subdural. Because the pressure in the venous system is much lower than in the arterial system, crescent-shaped subdural masses of blood often develop slowly before manifestations arise (e.g., hours-to-weeks). Chronic subdural hematoma is a risk in older persons, even after minor head trauma, with presentation often being ascribed to other conditions (e.g., Alzheimer's disease).

Subdural Hemorrhage

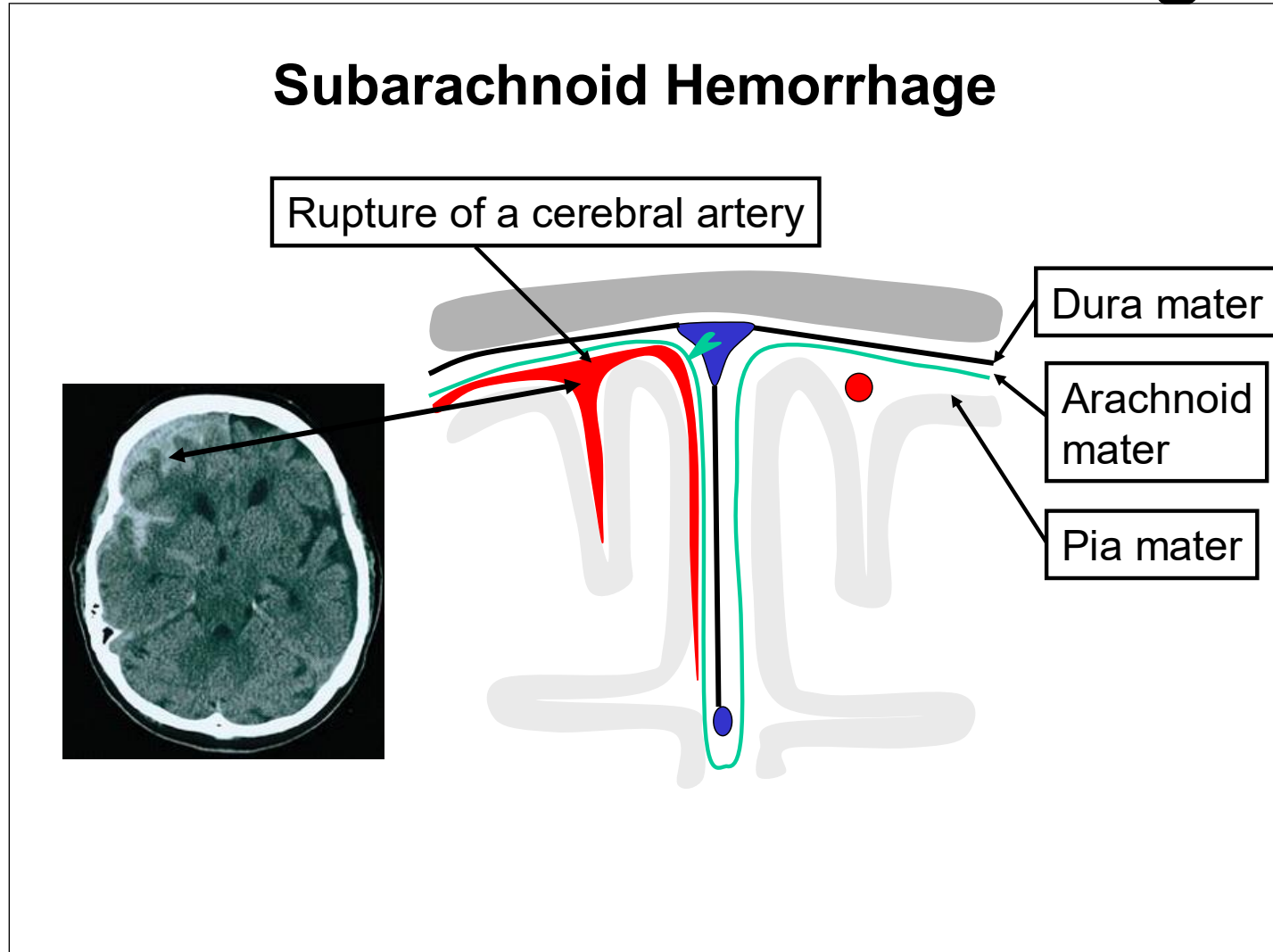
Subdural Hemorrhage



- Note crescent-shaped hematoma

- Many arteries reside between the arachnoid and pia mater (surrounded by CSF within the subarachnoid space). These vessels may rupture with trauma or spontaneously (e.g., in cases of aneurysm). The leaking blood assumes a tree-like appearance, while following sulci and other contours of the subarachnoid space. Intense headache, stiff neck and photophobia due to meningeal irritation are characteristic of subarachnoid hemorrhage.

Subarachnoid Hemorrhage



- Note blood-filled subarachnoid space

- Intracerebral hemorrhages reflect rupture of intraparenchymal branches of subarachnoid arteries, such as the lenticulostriate arteries (indicated by black box), which branch from the middle cerebral artery, supplying the internal capsule and basal ganglia.

Intracerebral Hemorrhage

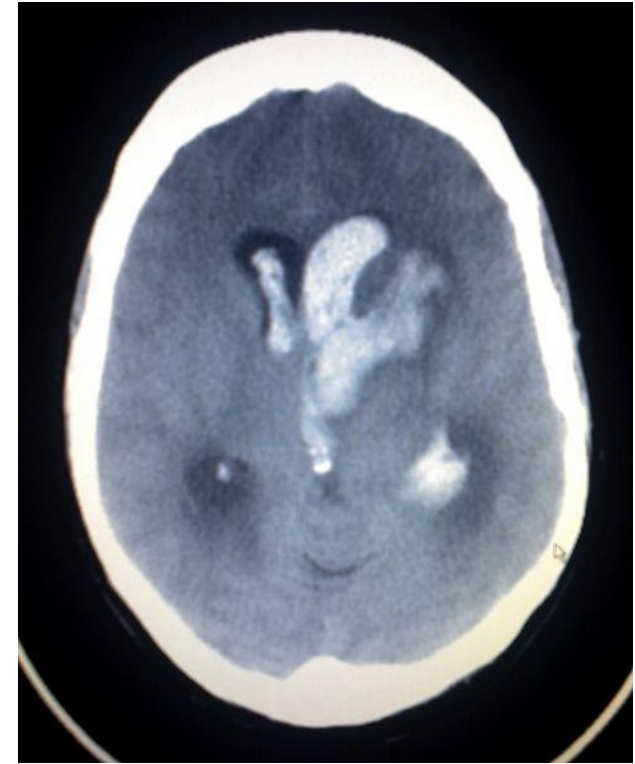
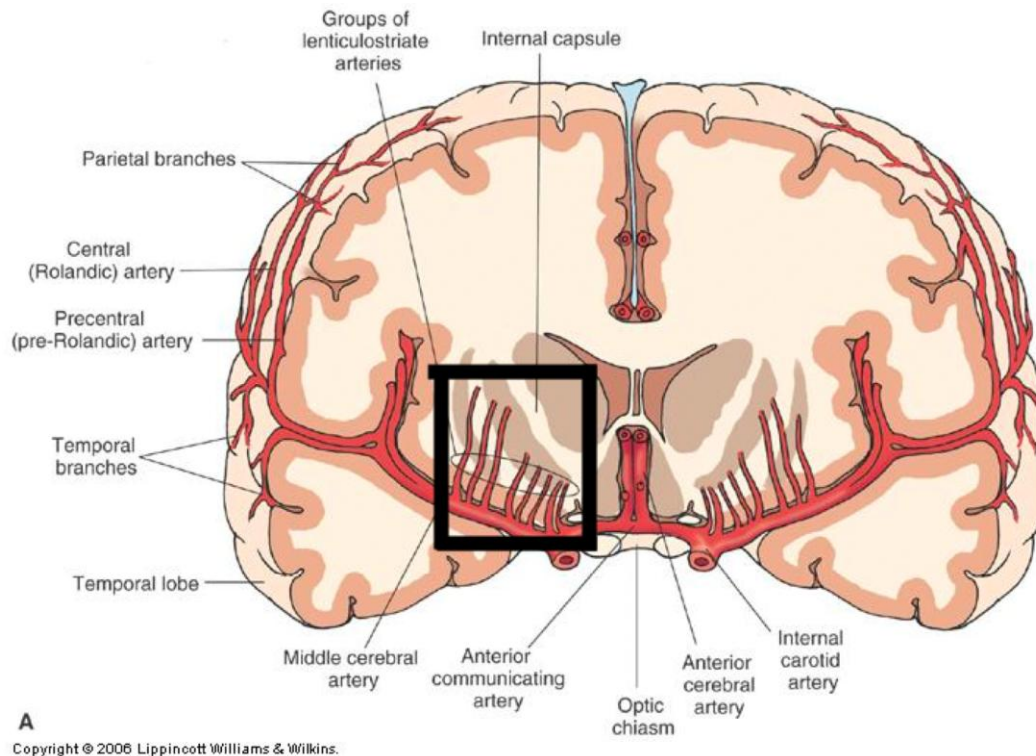


Figure 26–4a, Siegel and Sapru, 2019 http://en.wikipedia.org/wiki/File:Intracerebral_heamorrhage.jpg

- Small penetrating branches of major arteries are vulnerable to rupture (ventricular system may fill with blood)
 - Lenticulostriate arteries, branching from, the middle cerebral artery, are particularly vulnerable
 - Basal ganglial and internal capsular damage is clinically devastating

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- ***Internal Carotid and Vertebral Arteries***
- The arteries travel independently until entering the cranium, wherein they interface directly or indirectly via the anastomotic arterial circle of Willis.
- The right common carotid artery generally arises from the brachiocephalic trunk, and the left commonly arises from the aortic arch. At the level of the thyroid cartilage, the common carotids bifurcate, yielding the external and internal carotid arteries. The internal carotid arteries ascend through the deep neck, the carotid canals of the petrous bones and the cavernous sinus. They emit the ophthalmic and posterior communicating arteries. They divide, terminating as the anterior and middle cerebral arteries.

- The vertebral arteries are the first branches of the subclavian arteries. They ascend through the transverse foramina of the upper 6 cervical vertebrae and enter the skull through the foramen magnum. They commonly emit the posterior inferior cerebellar arteries (PICA), which usually spawn the posterior spinal arteries. The vertebral arteries then emit the anterior spinal arteries before unifying to form the basilar artery.

- The basilar artery gives rise to the anterior inferior cerebellar arteries (AICA). The superior cerebellar arteries arise from the basilar artery, just prior to its bifurcation into the two posterior cerebral arteries at the midbrain.
- The arterial circle of Willis is formed by a group of arteries that surround the optic chiasm, optic tract, mammillary bodies and the remaining ventral hypothalamus. Their perforating branches supply deep structures.

Cerebral Arteries

- Vertebral and internal carotid arteries anastomose at the basal diencephalon (by forming the circle of Willis)
- The vertebral arteries supply the basilar, cerebellar and cerebral arteries

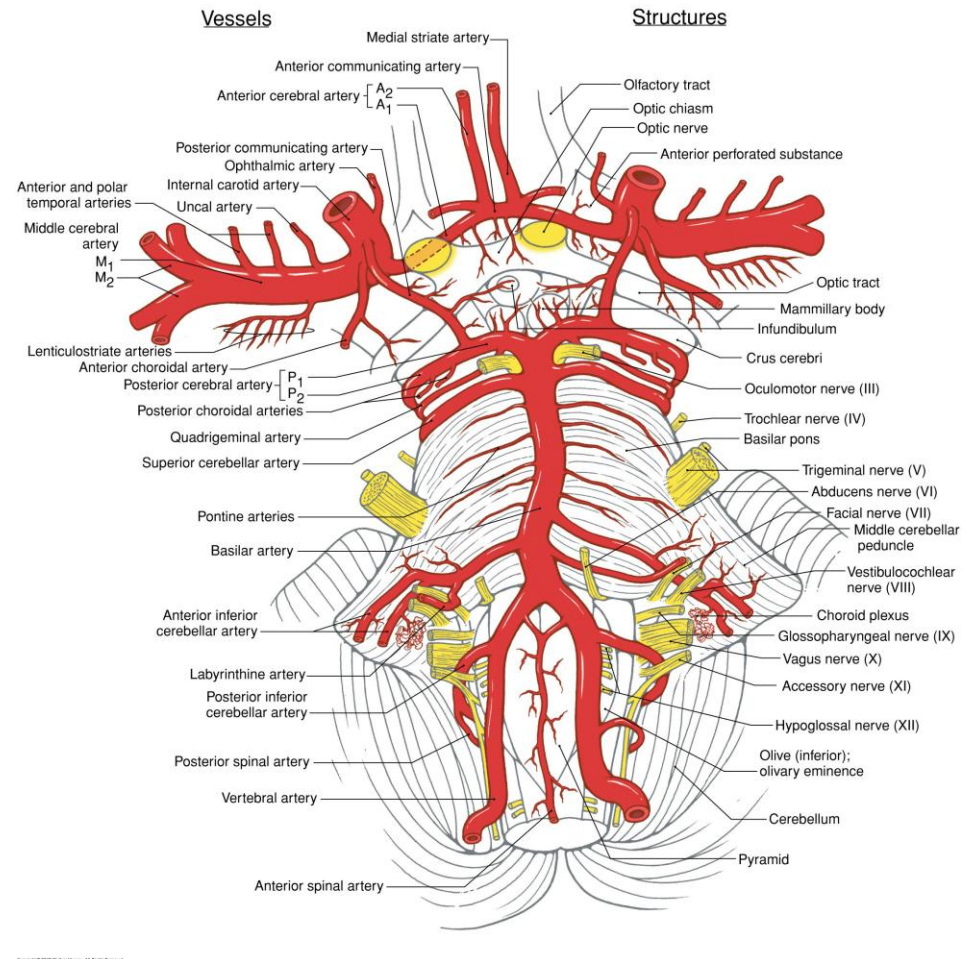


Fig. 2-21, Haines, 2024

- Anterior and middle cerebral arteries are the terminal divisions of the internal carotid artery, which reaches the ventral surface of the brain lateral to the optic chiasm. Arising from the junction of the internal carotid and middle cerebral arteries is the posterior communicating artery, which will join the vertebral arterial circulation.

- The anterior cerebral artery runs medially and anterior to the optic chiasm. Bridging the two anterior cerebral arteries is the anterior communicating artery.
- Major branches of the anterior cerebral arteries pass to the medial aspect of the frontal and parietal lobes, anterior perforated substance, septum pellucidum, and corpus callosum (often feeding all but the most posterior zones of this commissure).

- Striate arteries (early branches of the anterior cerebral artery) supply the basal ganglia and anterior limb of the internal capsule.
- The middle cerebral artery extends laterally in lateral fissure over the insula, sending branches to the lateral aspect of frontal, temporal, and parietal lobes. During its early course, the middle cerebral artery emits the lenticulostriate arteries, which target the basal ganglia and the entire internal capsule. These arteries are particularly vulnerable to rupture and are a major source of intracerebral (periventricular) hemorrhage.

Anterior and Middle Cerebral Arteries

- Internal carotid arteries feed the middle and anterior cerebral arteries
- Middle cerebral arteries emit the lenticulostriate arteries
- The anterior communicating artery connects the anterior cerebral arteries

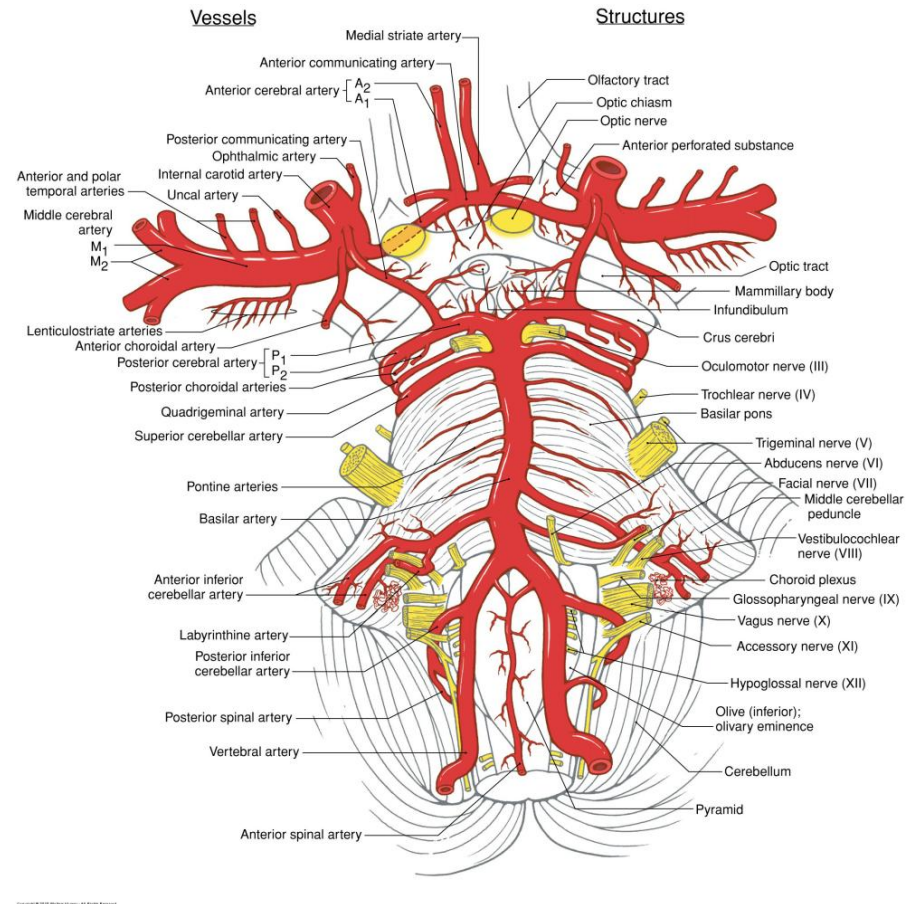


Fig. 2-21, Haines, 2024

- The posterior cerebral arteries are the terminal branches of the basilar artery. They receive the posterior communicating arteries from the internal carotid. They travel laterally and dorsally around the midbrain to reach the thalamus and third ventricle. Cerebral branches supply medial temporal, occipital, and parietal structures.

Posterior Cerebral Arteries

- Posterior cerebral arteries arise from the basilar artery, anastomosing with the internal carotid system via the posterior communicating arteries

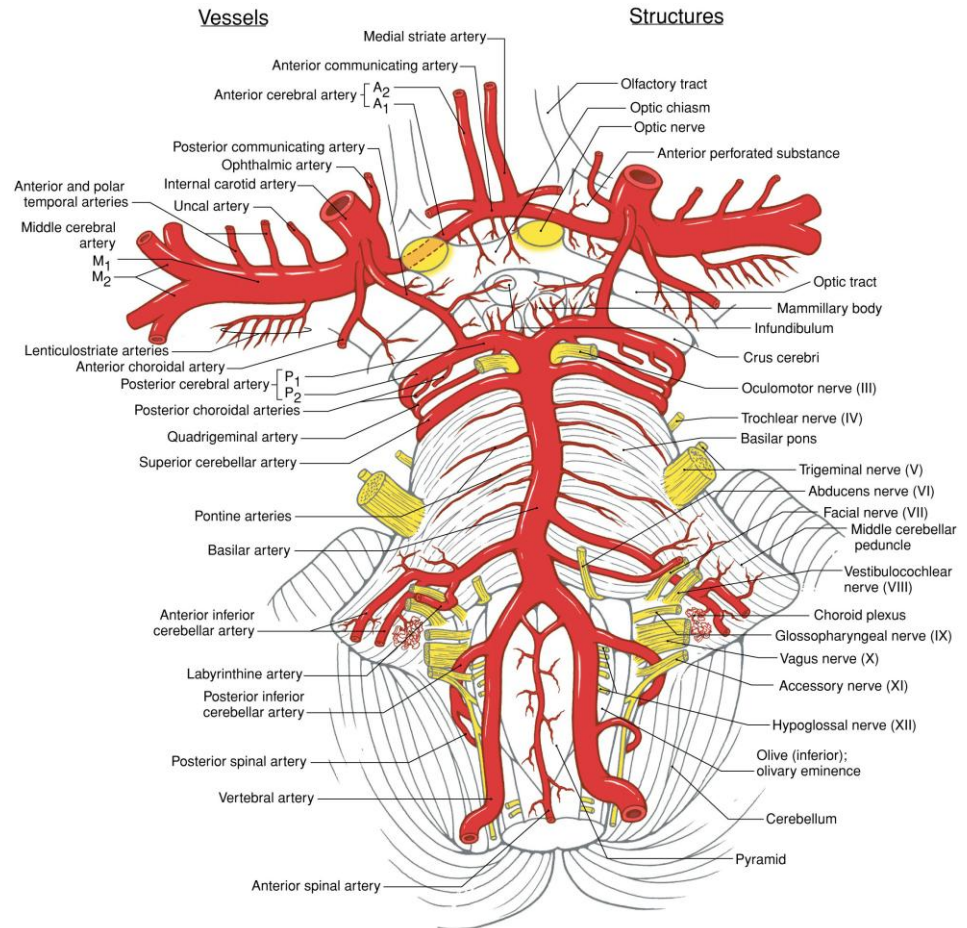
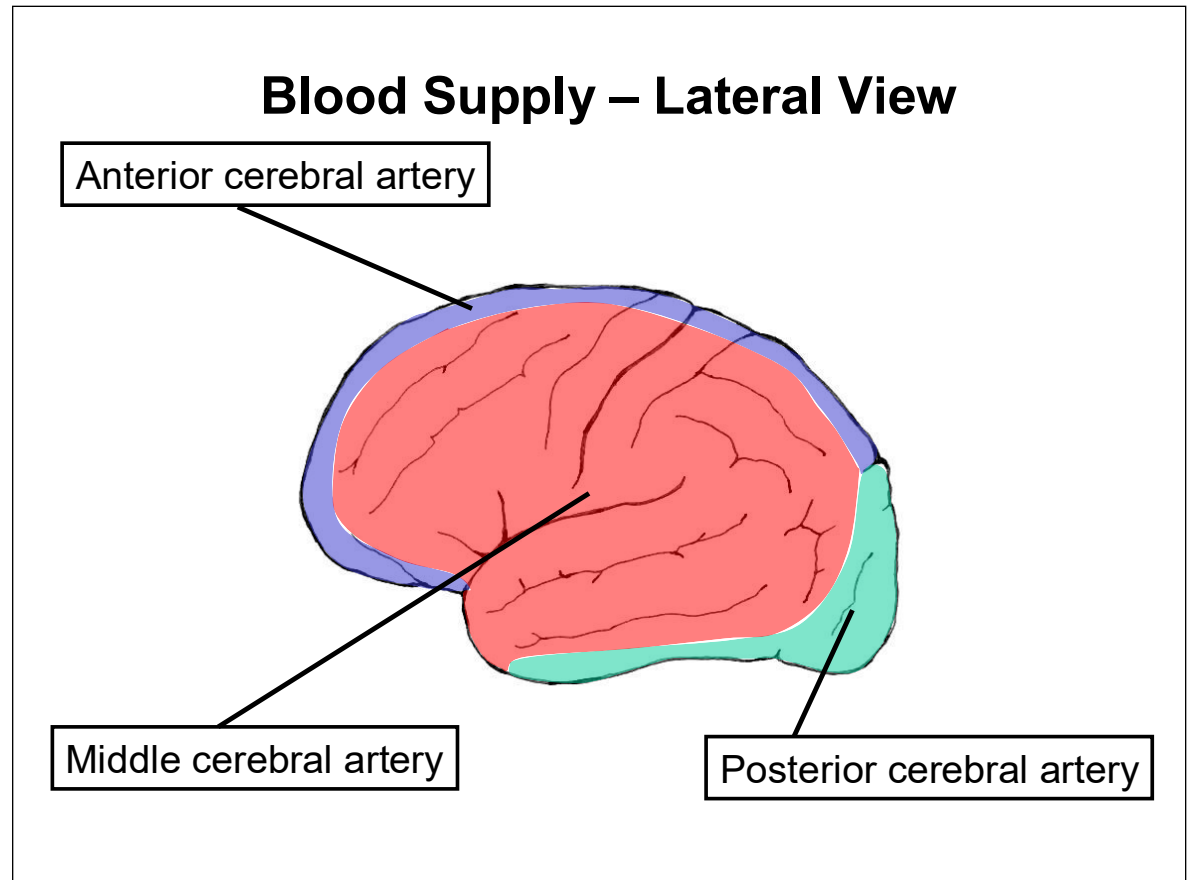


Figure 2–21, Haines, 2024

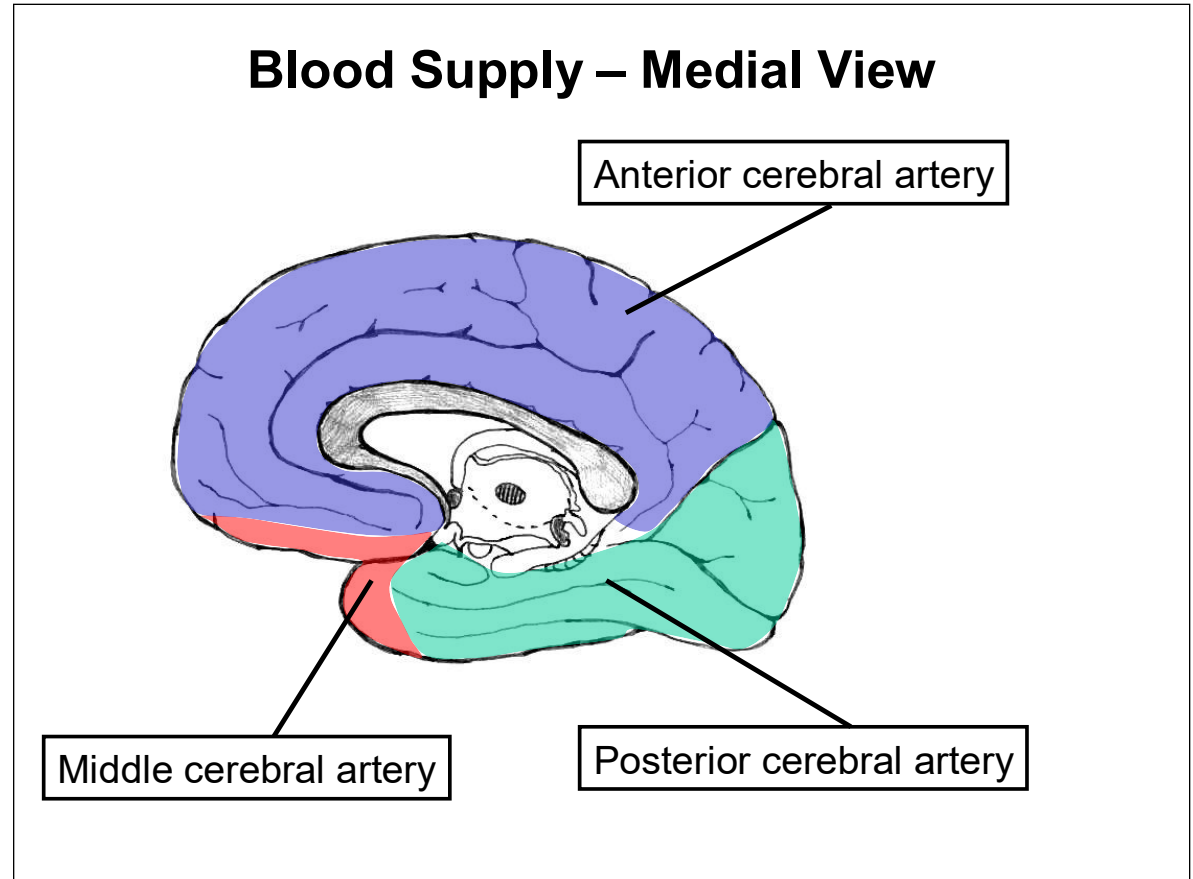
Cerebral Arteries: Lateral Distribution

- The middle cerebral artery dominates the lateral cerebral surface



Cerebral Arteries: Medial Distribution

- The anterior cerebral artery feeds the anteromedial cerebral surface
- The posterior cerebral artery feeds the posteromedial cerebral surface

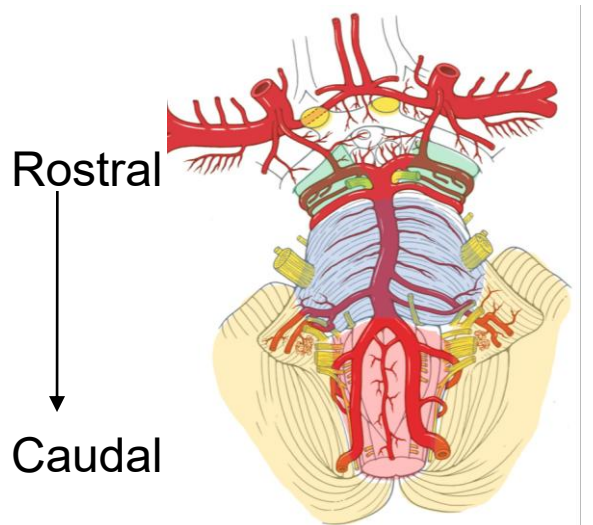


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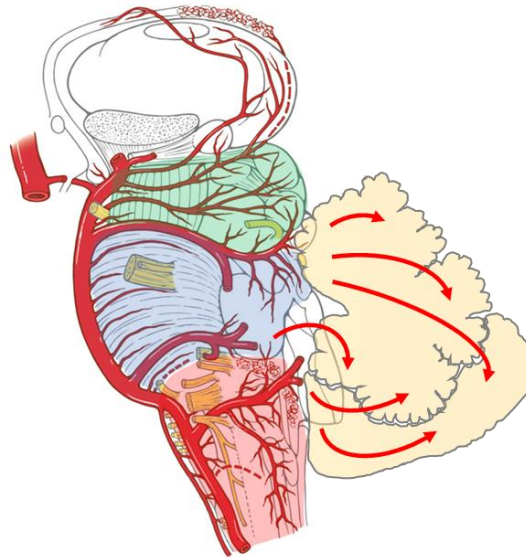
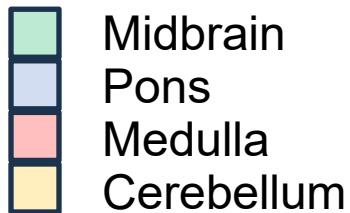
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- Blood typically reaches the brainstem, cerebellum and even parts of the diencephalon via the (subarachnoid) vertebral arteries. Branches from the vertebral arteries then distribute the bulbar blood, converging at the caudal pons to form the midline basilar artery. The vertebral arteries will supply blood to the cerebellum via three branches.

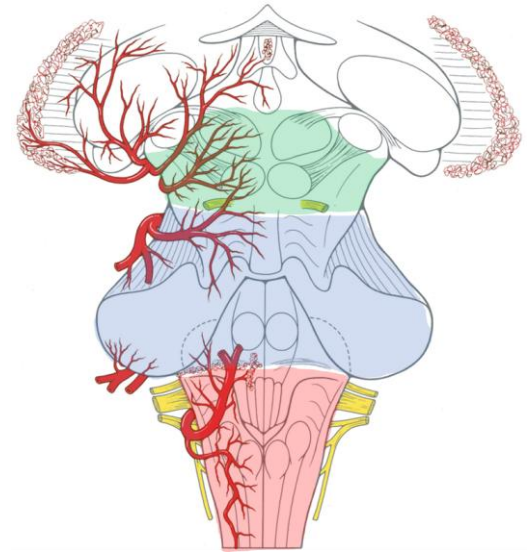
Bulbar Blood Supply



Ventral surface

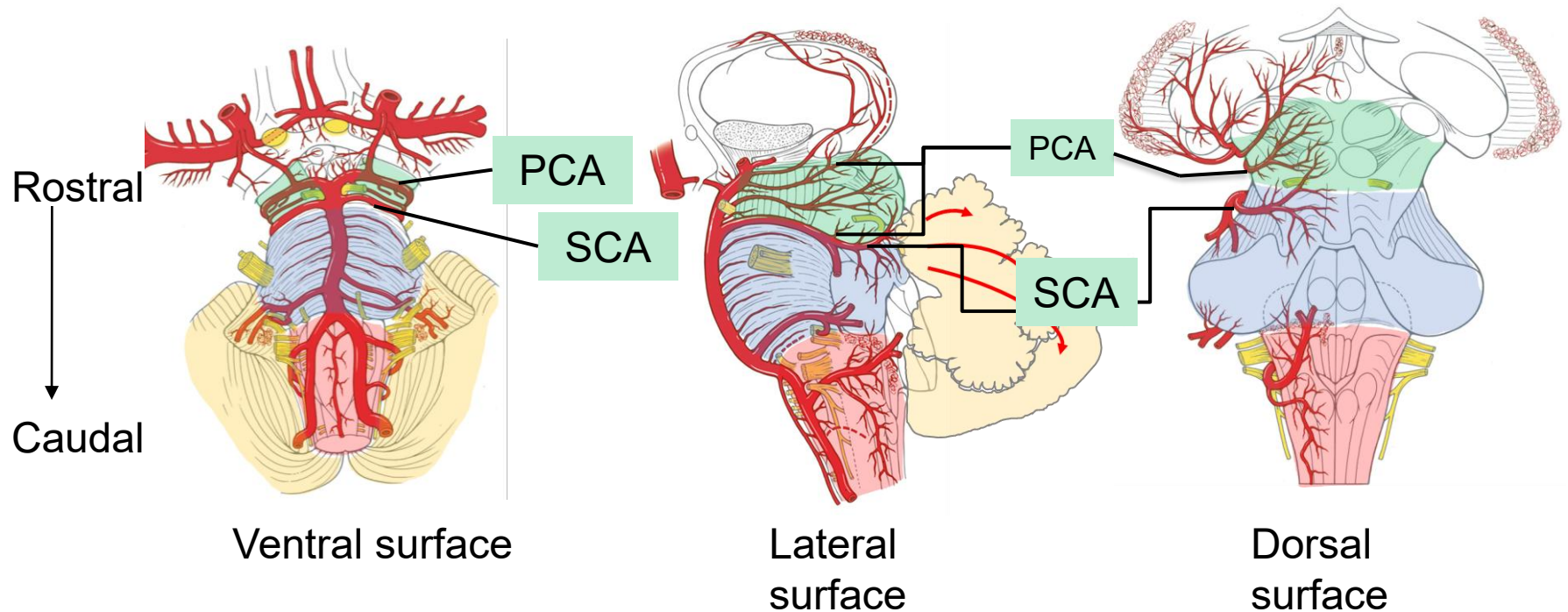


Lateral surface



Dorsal surface

Bulbar Blood Supply - Midbrain



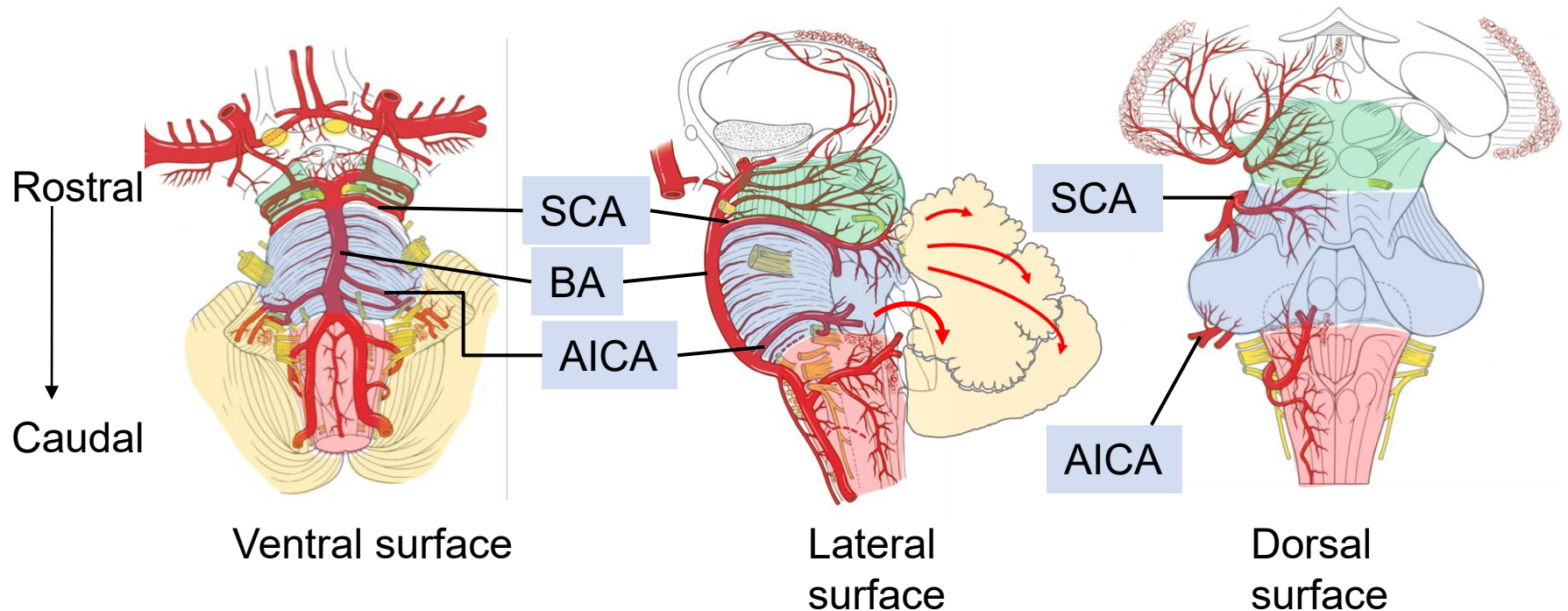
Midbrain receives supply from posterior cerebral artery (PCA) and superior cerebellar artery (SCA).

Ventral midbrain receives supply from PCA. Dorsal midbrain receives supply from SCA.

NSCI.0072

See also Haines, *Neuroanatomy Atlas 11e*, © Wolters Kluwer, **Figures 2-24, 2-21, 2-32**

Bulbar Blood Supply - Pons

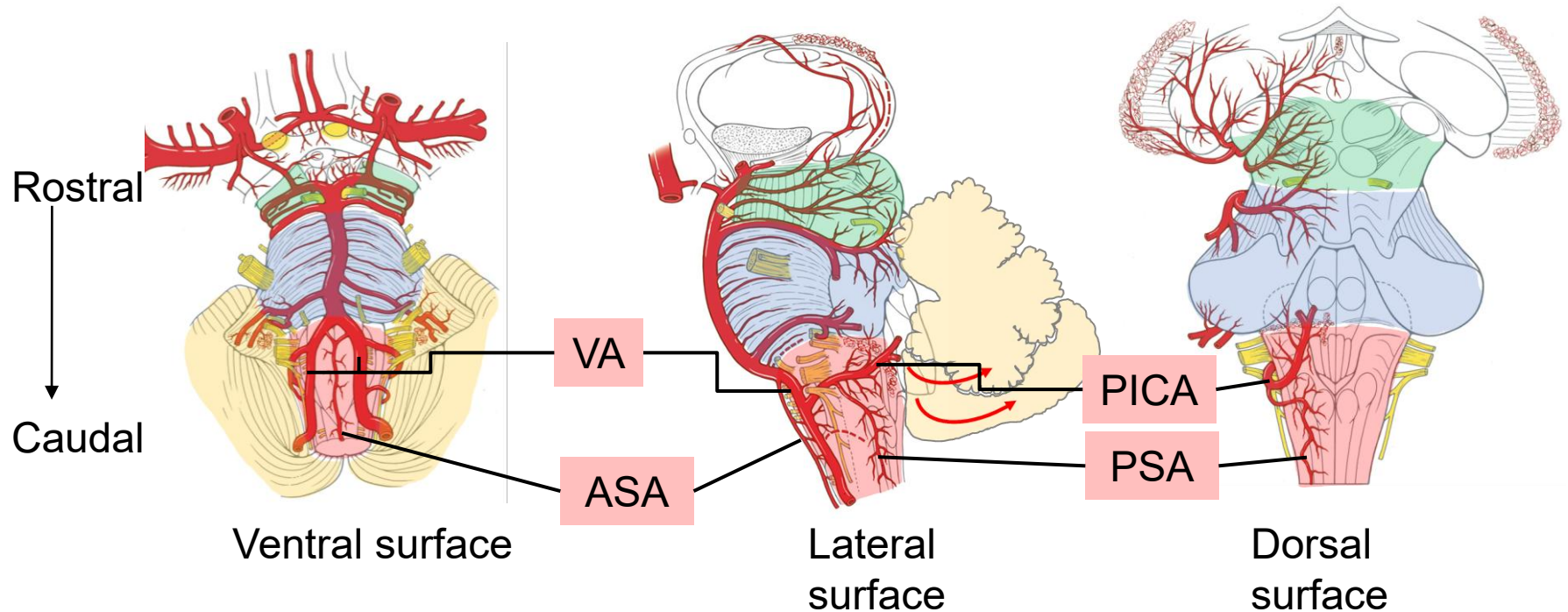


Pons receives supply from superior cerebellar artery (SCA), basilar artery (BA) and anterior inferior cerebellar artery (AICA).

Ventromedial pons receives supply from BA. Rostral dorsal pons (and part of cerebellum) receives supply from SCA. Caudal, dorsolateral pons (and parts of cerebellum) receives supply from AICA.

NSCI.0072

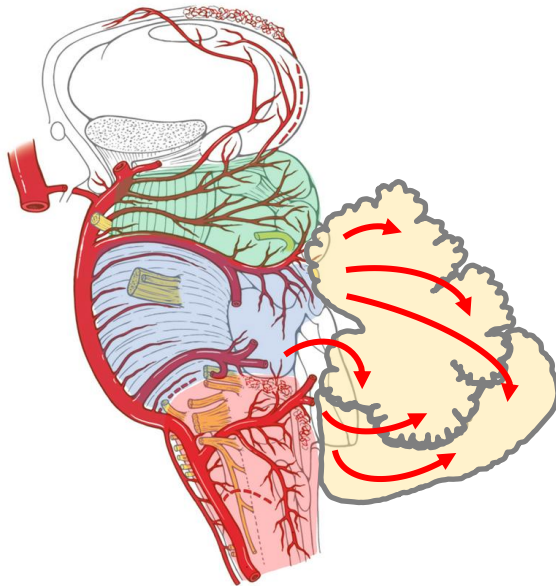
Bulbar Blood Supply - Medulla



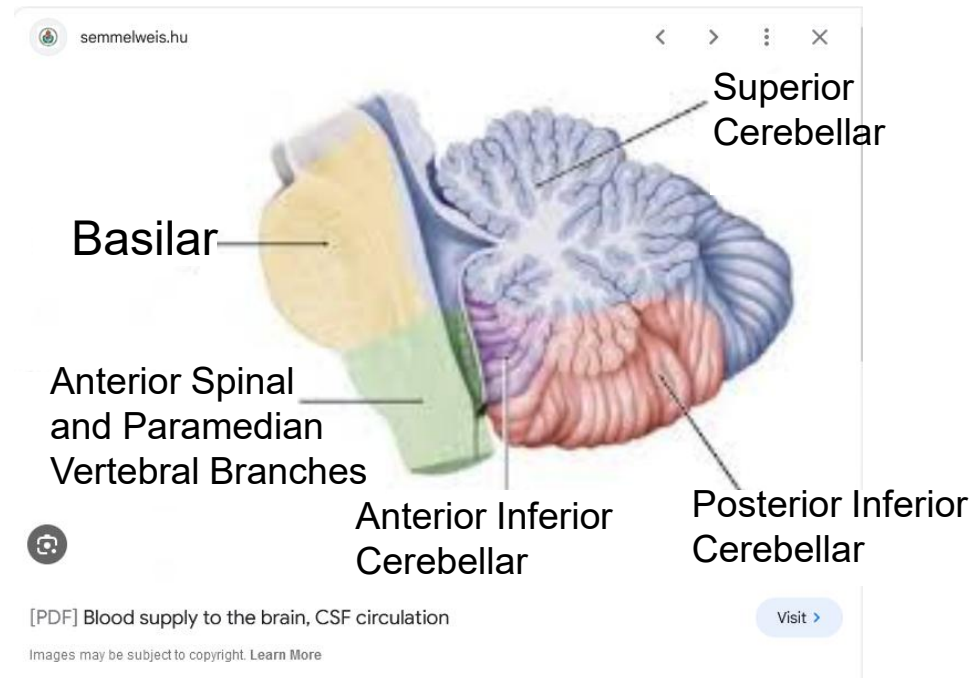
- Medulla receives supply from anterior spinal artery (ASA), posterior inferior cerebellar artery (PICA), posterior spinal artery (PSA) and vertebral artery (VA). Ventromedial medulla receives supply from ASA. Rostral, dorsolateral medulla (and parts of cerebellum) receives supply from PICA. Caudal dorsal medulla receives supply from PSA.

NSCI.0072

Blood Supply - Cerebellum



Lateral
surface



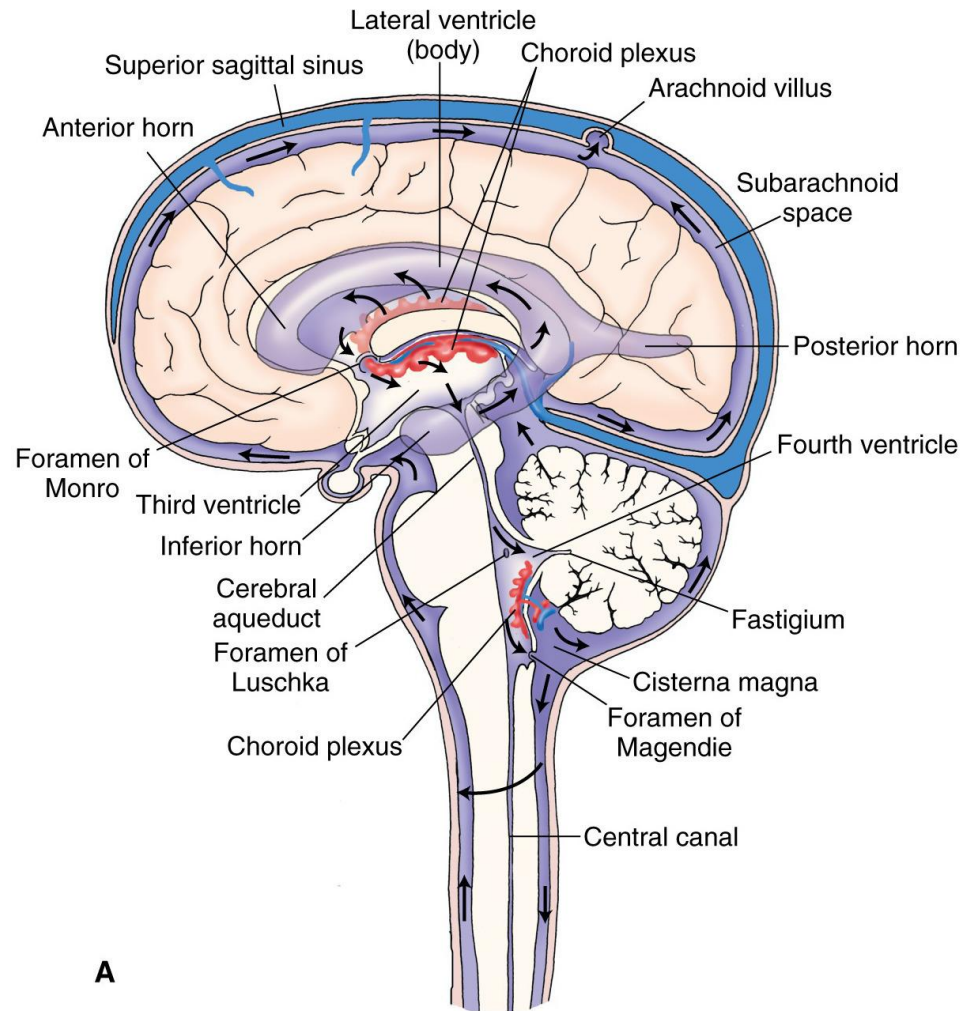
See also Haines, *Neuroanatomy Atlas 11e*, © Wolters Kluwer, **Figure 2-32**

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Spinal Meninges

- Spinal meninges lack the periosteal dura

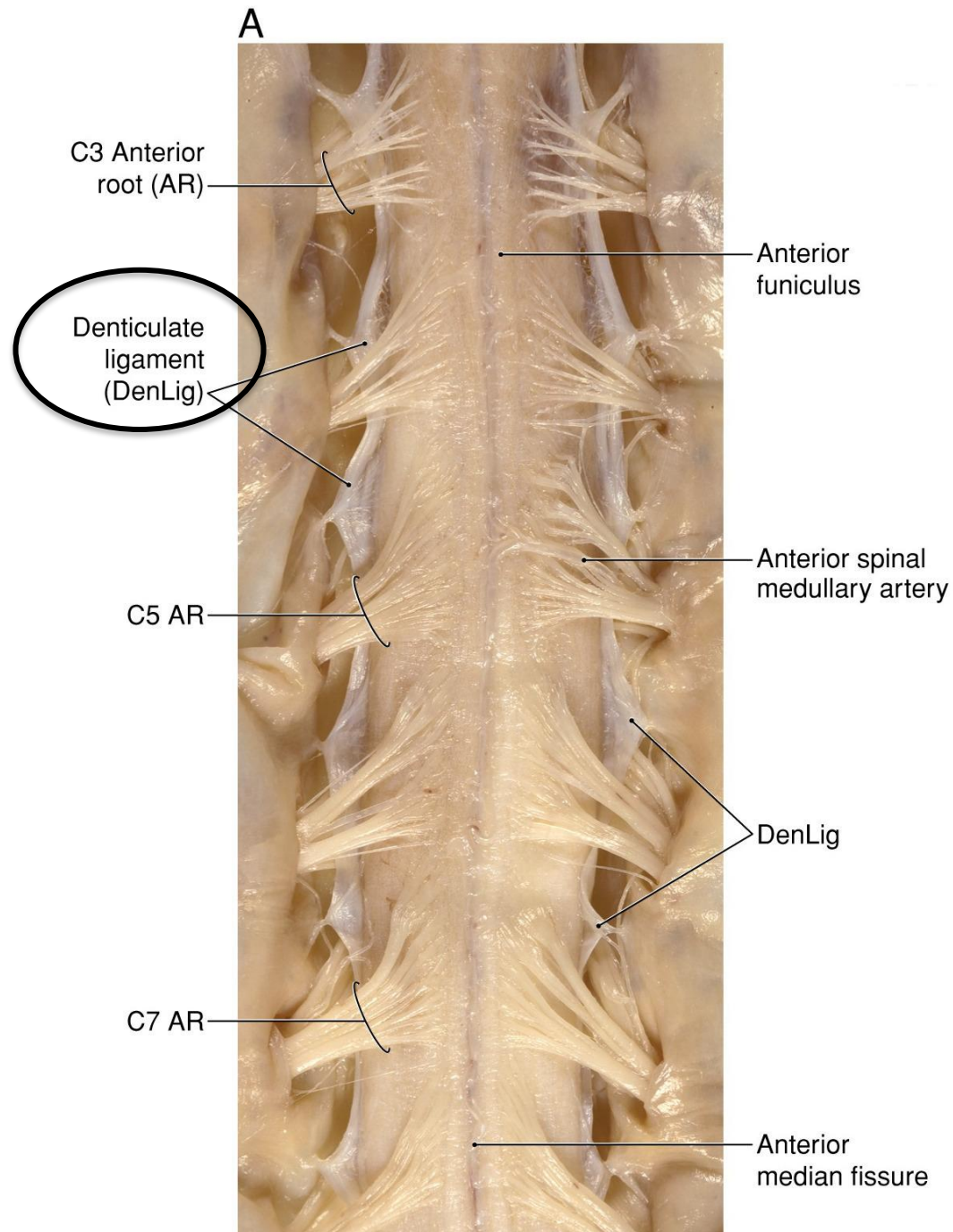


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Spinal Meninges

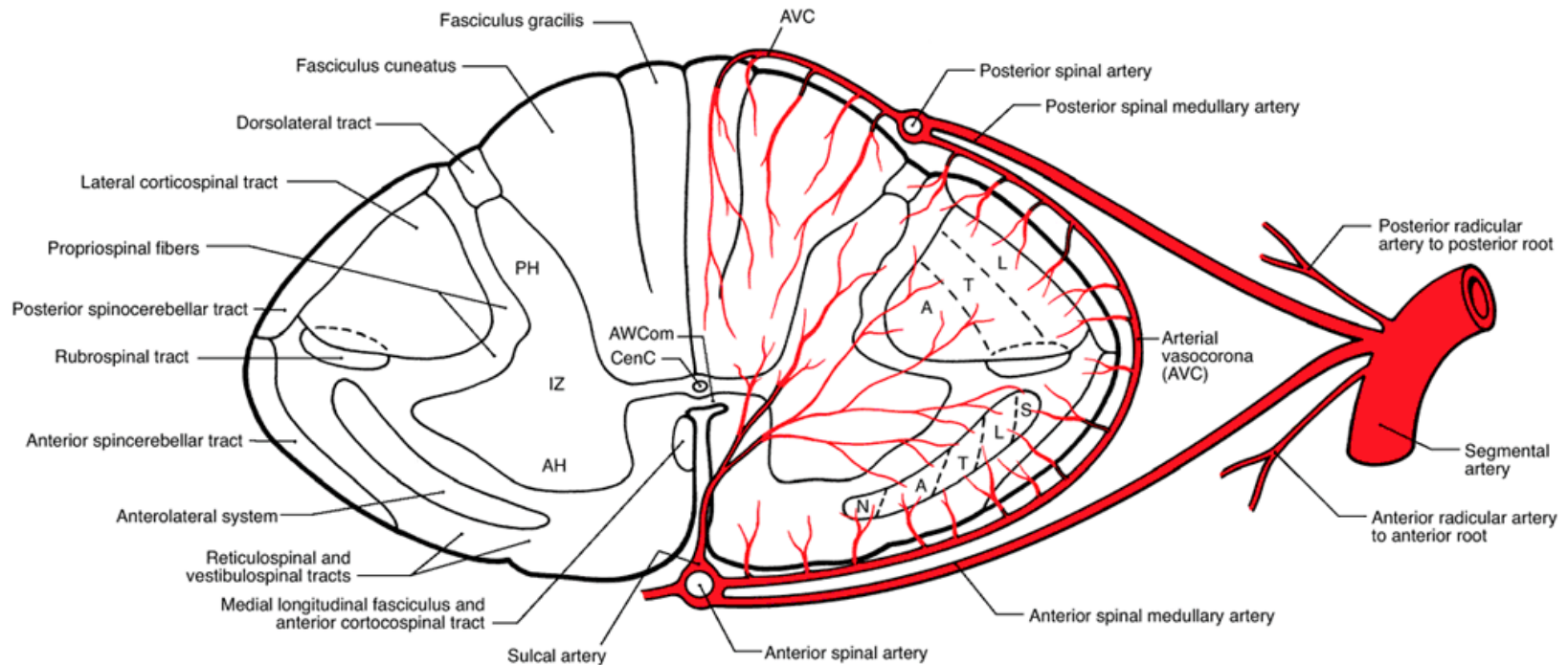
- Lateral extensions of the pia form denticulate ligaments, which fuse with the arachnoid and dura, lending stability



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Arterial Supply of the Spinal Cord



- Segmental branches of the vertebral arteries and aorta supply the anterior spinal and a pair of posterior spinal arteries via spinal medullary arteries
- Spinal anastomoses are rich and involve an arterial vasocorona
- Radicular arteries may also contribute, following nerve roots to the cord